GLUEING ALONG A WELL-ORDERED INCREASING UNION OF OPENS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Uniqueness of isomorphisms of solutions. Cohomology of Sheaves, Lemma cohomology-lemma-solution-in-finite-case. Two solutions of a system with vanishing negative exts are uniquely isomorphic.
- Filtered colimits are exact. Modules, Section modules-section-colimits. Stalk computation of the colimit complex.

Source reference. Original lemma 0D6A: sources/stacks-chow/tags.txt, source lines 16515–16515. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a ringed space. Let E be a well ordered set and let*

$$X = \bigcup\nolimits_{\alpha \in E} W_\alpha$$

be an open covering with $W_\alpha \subset W_{\alpha+1}$ and $W_\alpha = \bigcup_{\beta<\alpha} W_\beta$ if α is not a successor. Let K_α be an object of $D(\mathcal{O}_{W_\alpha})$ with $\mathrm{Ext}^i(K_\alpha, K_\alpha) = 0$ for $i < 0$. Assume given isomorphisms $\rho^\alpha_\beta : K_\alpha|_{W_\beta} \to K_\beta$ in $D(\mathcal{O}_{W_\beta})$ for all $\beta < \alpha$ with $\rho^\alpha_\gamma = \rho^\beta_\gamma \circ \rho^\alpha_\beta|_{W_\gamma}$ for $\gamma < \beta < \alpha$. Then there exists an object K in $D(\mathcal{O}_X)$ and isomorphisms $K|_{W_\alpha} \to K_\alpha$ for $\alpha \in E$ compatible with the isomorphisms ρ^α_β.

Proof. In this proof $\alpha, \beta, \gamma, \ldots$ represent elements of E. Choose a K-injective complex $I^\bullet_\alpha$ on W_α representing K_α. For $\beta < \alpha$ denote $j_{\beta,\alpha} : W_\beta \to W_\alpha$ the inclusion morphism. Using transfinite recursion we will construct for all $\beta < \alpha$ a map of complexes

$$\tau_{\beta,\alpha} : (j_{\beta,\alpha})_! I^\bullet_\beta \longrightarrow I^\bullet_\alpha$$

representing the adjoint to the inverse of the isomorphism $\rho^\alpha_\beta : K_\alpha|_{W_\beta} \to K_\beta$. Moreover, we will do this in such that for $\gamma < \beta < \alpha$ we have

$$\tau_{\gamma,\alpha} = \tau_{\beta,\alpha} \circ (j_{\beta,\alpha})_! \tau_{\gamma,\beta}$$

as maps of complexes. Namely, suppose already given $\tau_{\gamma,\beta}$ composing correctly for all $\gamma < \beta < \alpha$. If $\alpha = \alpha' + 1$ is a successor, then we choose any map of complexes

$$(j_{\alpha',\alpha})_! I^\bullet_{\alpha'} \to I^\bullet_\alpha$$

which is adjoint to the inverse of the isomorphism $\rho^\alpha_{\alpha'} : K_\alpha|_{W_{\alpha'}} \to K_{\alpha'}$ (possible because $I^\bullet_\alpha$ is K-injective) and for any $\beta < \alpha'$ we set

$$\tau_{\beta,\alpha} = \tau_{\alpha',\alpha} \circ (j_{\alpha',\alpha})_! \tau_{\beta,\alpha'}$$

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If α is not a successor, then we can consider the complex on W_α given by

$$C^\bullet = \text{colim}_{\beta<\alpha}(j_{\beta,\alpha})_! I_\beta^\bullet$$

(termwise colimit) where the transition maps of the sequence are given by the maps $\tau_{\beta',\beta}$ for $\beta' < \beta < \alpha$. We claim that $C^\bullet$ represents K_α. Namely, for $\beta < \alpha$ the restriction of the coprojection $(j_{\beta,\alpha})_! I_\beta^\bullet \to C^\bullet$ gives a map

$$\sigma_\beta : I_\beta^\bullet \longrightarrow C^\bullet|_{W_\beta}$$

which is a quasi-isomorphism: if $x \in W_\beta$ then looking at stalks we get

$$(C^\bullet)_x = \text{colim}_{\beta'<\alpha} \left((j_{\beta',\alpha})_! I_{\beta'}^\bullet\right)_x = \text{colim}_{\beta\leq\beta'<\alpha}(I_{\beta'}^\bullet)_x \longleftarrow (I_\beta^\bullet)_x$$

which is a quasi-isomorphism. Here we used that taking stalks commutes with colimits, that filtered colimits are exact, and that the maps $(I_\beta^\bullet)_x \to (I_{\beta'}^\bullet)_x$ are quasi-isomorphisms for $\beta \leq \beta' < \alpha$. Hence $(C^\bullet, \sigma_\beta^{-1})$ is a solution to the system $(\{K_\beta\}_{\beta<\alpha}, \{\rho_{\beta'}^\beta\}_{\beta'<\beta<\alpha})$. Since $(K_\alpha, \rho_\beta^\alpha)$ is another solution we obtain a unique isomorphism $\sigma : K_\alpha \to C^\bullet$ in $D(\mathcal{O}_{W_\alpha})$ compatible with all our maps, see Lemma 0D6A (this is where we use the vanishing of negative ext groups). Choose a morphism $\tau : C^\bullet \to I_\alpha^\bullet$ of complexes representing σ. Then we set

$$\tau_{\beta,\alpha} = \tau|_{W_\beta} \circ \sigma_\beta$$

to get the desired maps. Finally, we take K to be the object of the derived category represented by the complex

$$K^\bullet = \text{colim}_{\alpha\in E}(W_\alpha \to X)_! I_\alpha^\bullet$$

where the transition maps are given by our carefully constructed maps $\tau_{\beta,\alpha}$ for $\beta < \alpha$. Arguing exactly as above we see that for all α the restriction of the coprojection determines an isomorphism

$$K|_{W_\alpha} \longrightarrow K_\alpha$$

compatible with the given maps ρ_β^α. □